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Decarbonization With Hybrid Geothermal

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Many organizations are working to decarbonize building heating and cooling systems through electrification. Common solutions are air-source heat pumps, electric boilers or duct reheat systems. Geothermal is often not considered because of cost or insufficient land. By integrating a geothermal borefield with a supplemental energy source, hybrid geothermal systems offer a cost-effective, energy-efficient solution for decarbonizing heating in both new and existing buildings.

Traditional geothermal heating and cooling systems circulate water or a water-antifreeze mixture through underground pipes and back into the building to HVAC equipment—and possibly domestic hot water equipment. Geothermal is highly efficient because the ground temperature stays relatively constant throughout the year, enabling efficient energy extraction with geothermal heat pumps at a much higher coefficient of performance than electric resistance heat. An additional advantage of geothermal is that the ground acts as an energy storage device, storing summer heat for extraction during the winter.

The following are a few advantages of using geothermal heating and cooling:

- High efficiency;
- Reduces or eliminates on-site fossil fuel consumption;
- Reduces or eliminates potable water use for cooling tower systems;
- Reduced maintenance;

- Longer equipment life than air source heat pumps; and

- Reduces peak electrical demand and load on the grid compared to air source or electric resistance.

Barriers to using geothermal energy include:

- Construction cost;
- Lack of familiarity with geothermal systems by design professionals;
- Land availability; and
- Owners and operators uncomfortable with geothermal.

Hybrid Heating System

In cold climates, where more heating is needed than cooling, the ground temperature can cool over time. Inlet water temperatures colder than the design temperature reduce the heat pump's capacity and efficiency and create the risk of freezing the working fluid while affecting the system's ability to heat the building. A hybrid heating system is designed to prevent

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these problems by ensuring the working fluid stays within the design operating temperatures. Hybrid geothermal heating systems also reduce the borefield size over traditional geothermal system.

Two basic types of hybrid geothermal heating systems exist: those designed to provide instantaneous heat to help with peak conditions and those designed to provide seasonal heat to prevent the ground temperature from decreasing over time.

A hybrid heating system with a boiler acts on demand and injects heat into the geothermal water upstream of the heat pumps. The boiler is activated when the return water temperature from the borefield is lower than that required by the heat pump equipment, or at least 10°F (5.6°C) above the fluid's freezing point. The boiler should always be placed to heat the geothermal water upstream of the heat pumps (*Figure 1*). This will ensure heat provided by the boiler will go to the heat pumps prior to returning to the borefield. The boiler should never be placed on the geothermal return to the borefield, where heat from the boiler would be partially absorbed by the ground before heat pumps can use it.

For existing buildings, the designer must verify electrical capacity is available for hybrid electric boilers. Overlooking this could lead to expensive electrical upgrades or the inability to use the electric boiler.

A hybrid heating system with a solar thermal collector is not instantaneous, but it injects heat into the system on a seasonal basis, such as in the fall and winter. This warms the ground during the heating season to prevent the borefield temperature from decreasing over time.

Hybrid Cooling System

A hybrid geothermal cooling system uses a cooling tower, dry cooler or closed-circuit cooling tower to provide cooling on demand, daily or seasonally (*Table 1*). Of the three options, the benefit of a cooling tower is that it provides the coldest water, but a heat exchanger and additional pumps are required to separate the open cooling tower water from the geothermal water. There is increased water pressure drop from the heat exchanger, water evaporation from the tower, and water treatment is required. The dry cooler (*Figure 2*) supplies the warmest water but does not evaporate any water or require a heat exchanger. The closed-circuit cooling tower evaporates water indirectly by spraying it on the outside of the coils, which provides

FIGURE 1 Basic hybrid heating system.

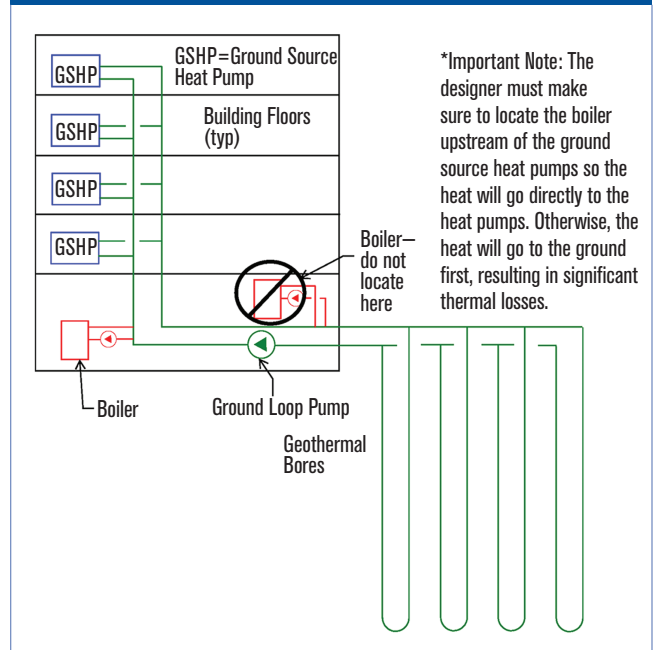


TABLE 1 Hybrid cooling options.

EQUIPMENT	EFFICIENCY	DESIGN COMPLEXITY	WATER CONSUMPTION	MAINTENANCE
Dry Cooler	Low	Low	None	Low
Closed Circuit Fluid Cooler	Medium	Medium	Low	Medium
Cooling Tower	High	High	High	High

FIGURE 2 Hybrid cooling system.

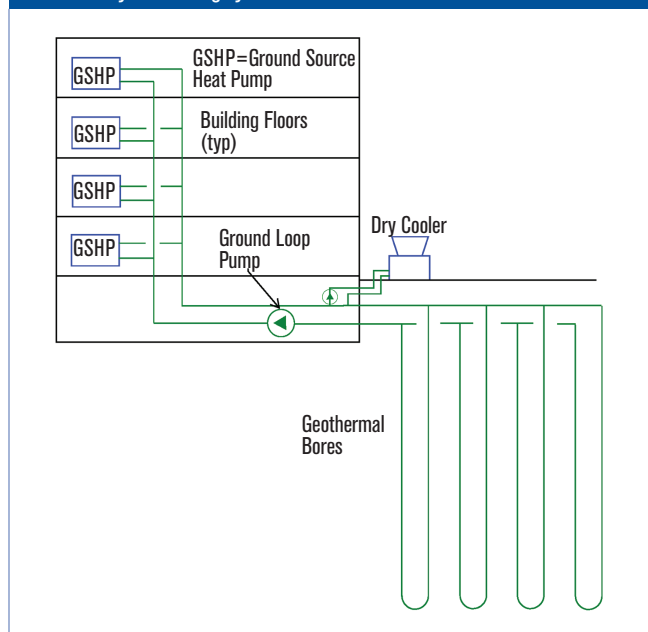


FIGURE 3 Traditional vs. hybrid geothermal borefield size.

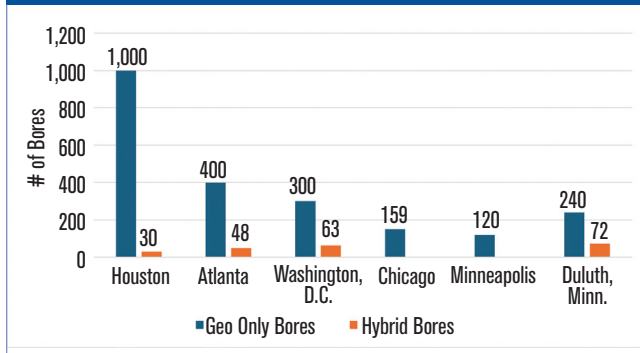


FIGURE 4 Building annual load comparison.

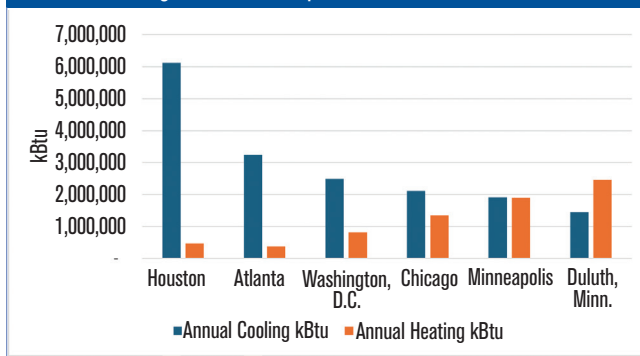
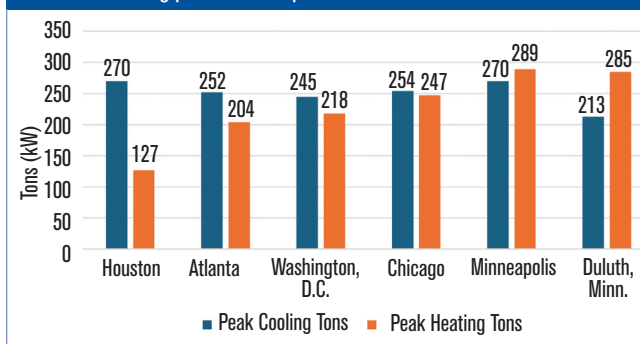


FIGURE 5 Building peak load comparison.



colder water than the dry cooler but warmer water than the cooling tower. Since the closed-circuit fluid cooler is a closed system, it does not require a heat exchanger.

The hybrid cooling system is activated when the outside air temperature is low enough to provide colder water from the hybrid cooling device than from the geothermal borefield. This improves the efficiency of the heat pumps and reduces the heat rejected to the ground. Care must be taken not to overcool the fluid when the system could switch to heating dominance. This can be done by only activating the hybrid system when the geothermal water is above a design setpoint.

A supplemental heat rejection device can precool the

ground at night for increased efficiency the next day or it can balance seasonal loads by operating at the end of winter or spring to cool the ground in preparation for summer.

Hybrid Geothermal Analysis

For this analysis, a prototypical four-story, 100,000 ft² (9290 m²) office building was modeled to meet ASHRAE Standard 90.1-2019¹ using eQuest. The borefield was sized (Figure 3) using a geothermal sizing software. The bores are 400 ft (122 m) deep with 1.25 in. (31.8 mm) diameter HDPE pipes, 1.0 Btu/h·ft·°F (1.7 W/m·K) or 1.2 Btu/h·ft·°F (2.1 W/m·K) grout, depending on the ground conductivity, and are spaced 20 ft (6.1 m) apart. (Designers note: The buildings evaluated in this study used 20 ft [6.1 m] borehole spacing to recognize limited land availability on some sites. While 20 ft [6.1 m] spacing is acceptable and commonly used where land is limited, wider spacing, typically 25 ft to 30 ft [7.6 m to 9.1 m] where feasible, is often preferred for improved thermal performance and decreased risk of drilling interference between bores.) The borefield was designed to stay within the temperature range of 30°F to 86°F (–1.1°C to 30°C) for a 20-year simulation. The hot climates used a dry cooler for supplemental heat rejection, and the cold climates used an electric boiler as needed. For hybrid systems, the borefield is sized for the nondominant load.

Houston, Atlanta and Washington, D.C., buildings were cooling dominant with the hybrid borefields sized to meet the heating load of the building. Chicago and Minneapolis buildings had relatively balanced heating and cooling loads and did not benefit from a hybrid geothermal system, but required antifreeze. Duluth was heating dominant with the hybrid borefield sized to meet the cooling load of the building.

Houston demonstrated the greatest performance disparity between traditional and hybrid geothermal systems, with Atlanta and Washington, D.C., showing the next most pronounced differences. In Duluth, where the ground temperature averages around 48°F (8.9°C), and annual heating loads exceed the cooling loads, the hybrid system was optimized by incorporating both a boiler and antifreeze.

A comparison between a building's annual heating and cooling loads and the size of the borefield (Figure 4)

shows that they are closely related, indicating that buildings with a large difference between the annual heating and cooling loads will require larger borefields.

A low correlation exists between the peak heating and cooling loads and the borefield size (*Figure 5*). A building can have similar peak heating and cooling loads but require a large borefield because of a large annual imbalance between heating and cooling loads.

Note: The Minneapolis and Houston peak cooling loads are the same because each building is designed to meet ASHRAE Standard 90.1-2019 for its location. In Houston, the Standard 90.1-2019 maximum window SHGC is 0.25, but in Minneapolis, it is 0.38, which allows more solar heat into the building.

A simple analogy on how borefield size is correlated to the annual heating and cooling loads is comparing the loads to water and the ground to a barrel. During the summer, you are symbolically filling the barrel with water; the next winter, you are taking the water back out. If an imbalance exists, and you put twice as much water in each summer as you take out the following winter, your barrel will eventually overflow. This is why a large barrel is needed with a traditional geothermal system with a large heating or cooling imbalance—so it doesn't overflow (overheat) or run dry (overcool).

With a hybrid system, you are in control of how much water you put into the barrel (ground). When it gets full, you stop adding water. For a cooling-dominant system, you size the barrel based on what you need to take out in the winter, knowing you will have plenty of water to fill it up again next summer.

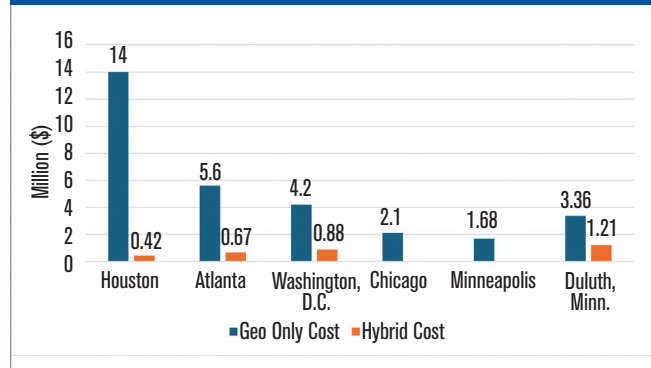
In reality, many other factors, such as groundwater movement and heat dissipation away from the borefield, will carry heat away from the site, but this is a simple analogy that helps illustrate the concept.

Cost Comparison

Hybrid geothermal systems can significantly decrease the initial cost. *Figure 6* is a borefield cost comparison between traditional and hybrid geothermal systems, including the auxiliary equipment.

The vertical bore cost is assumed to be \$35/ft (\$115/m) for all locations, but this cost can vary depending on location, and local drillers should be consulted for pricing. For the hybrid cooling systems, 2024 construction cost estimating software was used to estimate the cost of the dry cooler, boiler, piping and

FIGURE 6 Traditional vs. hybrid geothermal cost.



controls. The antifreeze concentration was 20% and was \$13/gallon (\$3.40/L) plus \$5,000 for the glycol feed tank. The building-level HVAC equipment was assumed to be the same for the hybrid and traditional options and is omitted from the cost comparison.

Using a hybrid system in Houston has the most significant impact since it has the most lopsided annual cooling vs. heating loads. As the annual loads become more balanced, the difference decreases. The dry cooler and boiler cost had little impact on the overall cost when compared to the borefield cost. Chicago and Minneapolis did not benefit from a hybrid system, so only the borefield option is shown.

Sizing a Hybrid Geothermal System

In this article, only water-to-air heat pumps are shown, which would always be sized based on the heating and cooling loads in the spaces served. If the building used a water-to-water heat pump along with boilers and chillers in parallel, the heat pumps would be sized for the nondominant peak load, and the supplemental chiller or boiler would be sized for the remaining load. For a cooling tower system, the cooling tower is sized to provide the difference between the peak cooling and heating loads.

An in-depth analysis of how to size and control a hybrid geothermal system is outside this article's scope. A couple helpful resources on this are "Development of Design Guidelines for Hybrid Ground-Coupled Heat Pump Systems (ASHRAE TRP-1384)"² and "Study on Operating and Control Strategies for Hybrid Ground Source Heat Pump System."³

Challenges with Hybrid Geothermal

A few challenges exist when using a hybrid

geothermal system compared to a traditional geothermal system. Hybrid systems have increased maintenance, equipment and control complexity. For these reasons, many times the ground heat exchanger size is increased instead of using a hybrid approach. The design team should compare the costs and benefits of each system.

Analysis Limitations

This analysis included only a typical office building. Traditional geothermal has been done successfully for years in K–12 applications in the southern U.S., and residential buildings have lower internal heat gains, which also results in more balanced loads for central and southern U.S. climates. Buildings with high internal heat gains such as data centers or manufacturing facilities can be cooling dominant, even in cold northern climates.

Each project needs to be analyzed with an hourly load model or an entire year's data to determine a hybrid system's benefit. Peak load data alone isn't sufficient for

proper geothermal system design and analysis.

Conclusion

In buildings with a large imbalance in annual heating and cooling loads, hybrid geothermal systems can decarbonize heating systems at a lower cost than traditional geothermal systems. Hybrid systems are more energy efficient than traditional electric resistance and air-source heat pumps, helping further reduce carbon footprints while maintaining economic viability. It can provide many of the benefits of geothermal energy at a lower cost and with less land use.

References

1. ANSI/ASHRAE/IES Standard 90.1—2019, *Energy Standard for Buildings Except Low-Rise Residential Buildings*.
2. Hackell, S., G. Nellis, S. Klein, J. Thornton. 2008. "Development of Design Guidelines for Hybrid Ground-Coupled Heat Pump Systems." ASHRAE Research Project RP-1384, Final Report.
3. Zhang, Q., N. Lv, C. Shengpeng, S. Hao, C. Zhenqian. 2015. "Study on operating and control strategies for hybrid ground source heat pump system." *Procedia Engineering* 121:1894–1901. <https://doi.org/10.1016/j.proeng.2015.09.172> ■

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